

# Open Models for Generative Artificial Intelligence

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<https://floss.social/@jgbarah>

<https://jgbarah.github.io/presentations>

Inteligencia Artificial: Actividades didácticas  
Madrid, Spain, May 19 2025



# Plan

- 1 What is an “open” model?
- 2 Generative AI in your computer
- 3 Infrastructure to play, to share
- 4 Many issues raised
- 5 Summarizing

Open Generative  
AI

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What is an “open”  
model?

Generative AI in  
your computer

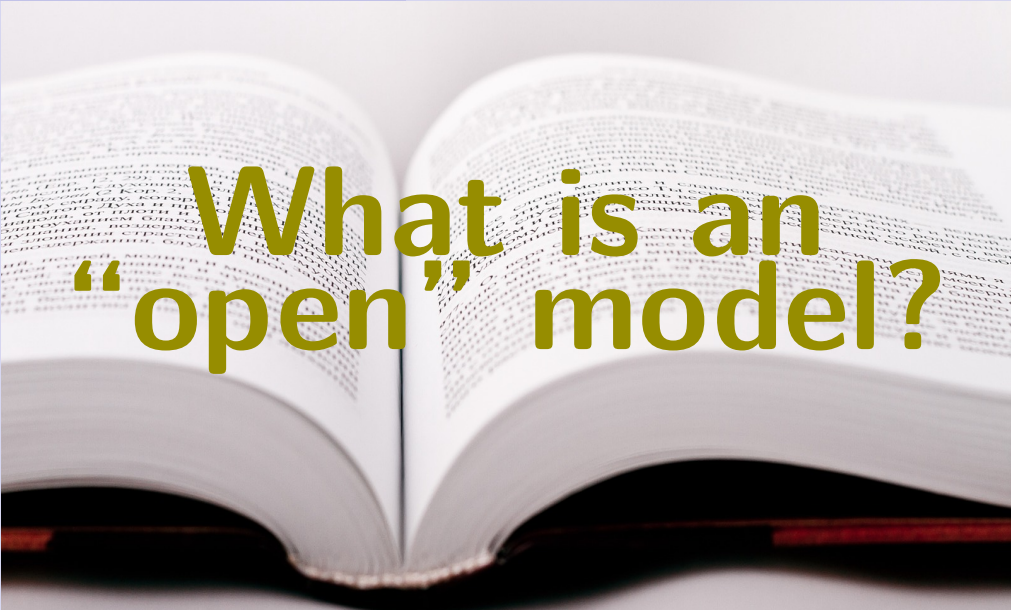
Infrastructure to  
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Many issues raised

Summarizing

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# What is an “open” model?



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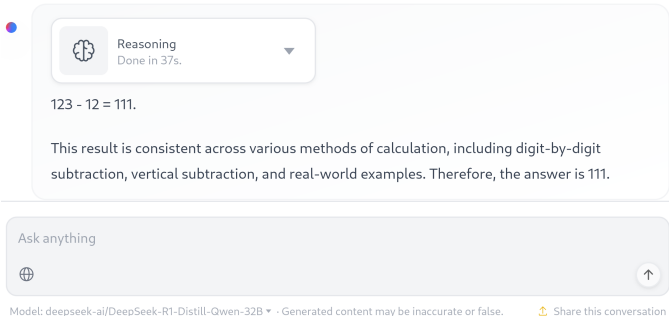
Many issues raised

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# Start by trying!

Dime cuánto es 123 menos 12



The screenshot shows a chat interface with a light blue background. At the top, there's a user prompt: "Dime cuánto es 123 menos 12". Below it, a response box contains a reasoning process. The response starts with a small icon of a brain in a square, followed by the text "Reasoning Done in 37s." and a dropdown arrow. Below this, the calculation "123 - 12 = 111." is shown. Further down, a paragraph explains: "This result is consistent across various methods of calculation, including digit-by-digit subtraction, vertical subtraction, and real-world examples. Therefore, the answer is 111." At the bottom of the chat area, there's a text input field with the placeholder "Ask anything" and a globe icon on the left, and an upward arrow icon on the right. Below the chat area, a footer line reads: "Model: deepseek-ai/DeepSeek-R1-Distill-Qwen-32B · Generated content may be inaccurate or false." followed by a share icon and the text "Share this conversation".

Reasoning  
Done in 37s.

123 - 12 = 111.

This result is consistent across various methods of calculation, including digit-by-digit subtraction, vertical subtraction, and real-world examples. Therefore, the answer is 111.

Ask anything

Model: deepseek-ai/DeepSeek-R1-Distill-Qwen-32B · Generated content may be inaccurate or false. [Share this conversation](#)

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# Start by trying!

<https://huggingface.co/chat/>

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# Models for text, images...



Stable Diffusion XL:  
selfie of a teacher  
with a dinosaur, in  
Manhattan  
cinematic-default

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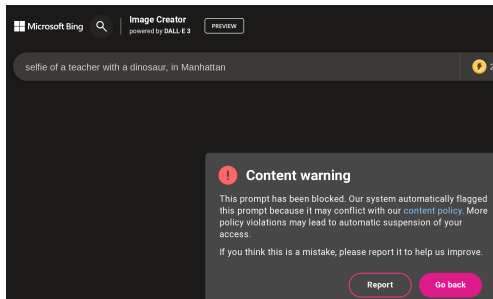
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# But beware!



“selfie of a teacher with a dinosaur, in Manhattan”

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# Can we control how we use generative models?

# What's a generative model

- Architecture
- Weights, just weights
- Software to make inferences

But also:

- Data to train, benchmark
- Software to train, benchmark
- Weights of intermediate models
- Documentation, explaining everything

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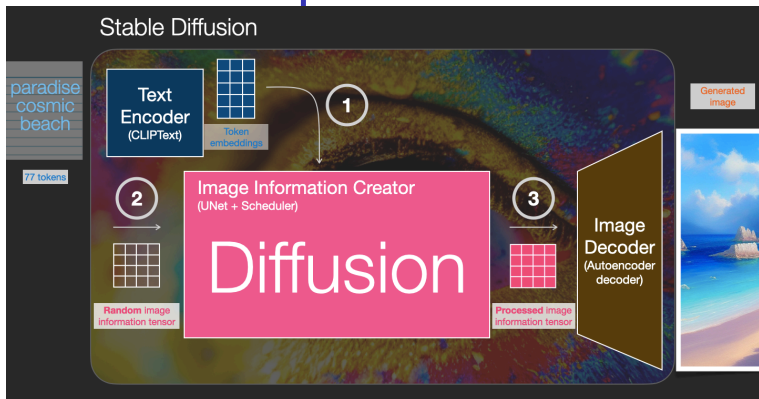
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# Example: Stable Diffusion



<https://jalammar.github.io/illustrated-stable-diffusion/>

---

# The Model Openness Framework: Promoting Completeness and Openness for Reproducibility, Transparency and Usability in AI

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**Matt White<sup>1,2</sup>, Ibrahim Haddad<sup>2</sup>, Cailean Osborne<sup>2,3</sup>,  
Xiao-Yang (Yanglet) Liu<sup>1,4</sup>, Ahmed Abdelmonsef<sup>1</sup>, Sachin Varghese<sup>1</sup>**

<sup>1</sup> LF AI & Data - Generative AI Commons, <sup>2</sup> Linux Foundation,

<sup>3</sup> University of Oxford, <sup>4</sup> Columbia University

<https://arxiv.org/abs/2403.13784>

# Classes of “openness”

- Open science: paper, datasets, log files, intermediate models parameters & metadata
- Open tooling: training, inference, and evaluation code, libraries, evaluation data
- Open model: architecture, model parameters & metadata, tech report, model card, data card

# “Open reproducible model”

All of the above, and

complete training set

(and any other detail needed to produce the same  
binary model)

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# For each of them...

- Freedom of use
- Freedom of study
- Freedom of modification
- Freedom of sharing

# Minimum for running in your computer

- Inference code, with libraries
- Model architecture
- Model parameters & metadata (weights)
- Technical report (prompts, etc.)
- Model card (convenient)

# Equipment requirements

- Depends on architecture, and size of the model
- CPU can be enough, GPU can accelerate a lot
- RAM enough so that model fits
- Code is usually FOSS

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# Example: HuggingChat

**HuggingChat** v0.8.4

Making the community's best AI chat models available to everyone.

**Current Model**  
**CohereForAI/c4ai-command-r-plus**  
[Model page](#) [Website](#)

**Examples**

- Write an email from bullet list
- Code a snake game
- Assist in a task

☐ Search web ⓘ

Ask anything

Model: CohereForAI/c4ai-command-r-plus • Generated content may be inaccurate or false.

**Settings**

Application Settings

**Models**

- CohereForAI/c4ai-command-r-plus** **Active**
- meta-llama/Meta-Llama-3-70B-Instruct
- HuggingFaceH4/zephyr-orpo-141b-A35b-v0.1
- mistralai/Mistral-8x7B-Instruct-v0.1
- NousResearch/Nous-Hermes-2-Mistral-8x7B-DPO
- google/gemma-1.1-7b-it

**CohereForAI/c4ai-command-r-plus**  
 Command R+ is Cohere's latest LLM and is the first open weight model to beat GPT4 in the Chatbot Arena

[Model page](#) [Model website](#)  
 Copy direct link to model

**Active model**

**System Prompt**

<https://huggingface.co/chat/>

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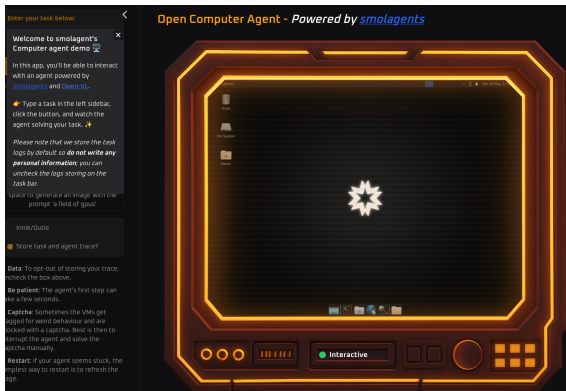
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# Example: computer-agent



<https://huggingface.co/spaces/smolagents/computer-agent>

# Example: computer-agent

## Key components:

- Agents framework: smolagents
- Model: Qwen2-VL-72B
- Desktop: E2B Desktop

<https://github.com/huggingface/smolagents>  
<https://huggingface.co/Qwen/Qwen2-VL-72B-Instruct>  
<https://github.com/e2b-dev/desktop>

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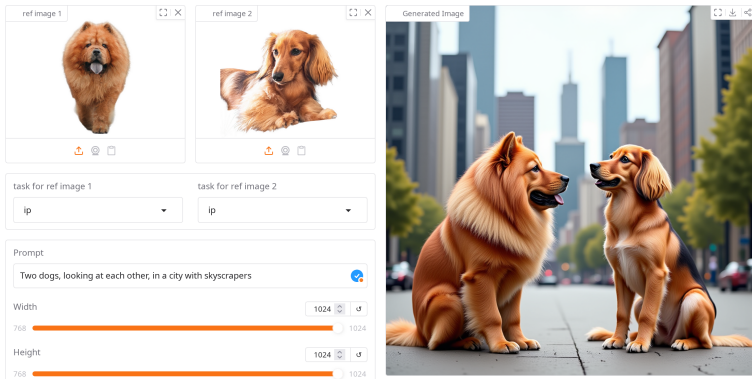
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# Example: DreamO



<https://huggingface.co/spaces/ByteDance/DreamO>

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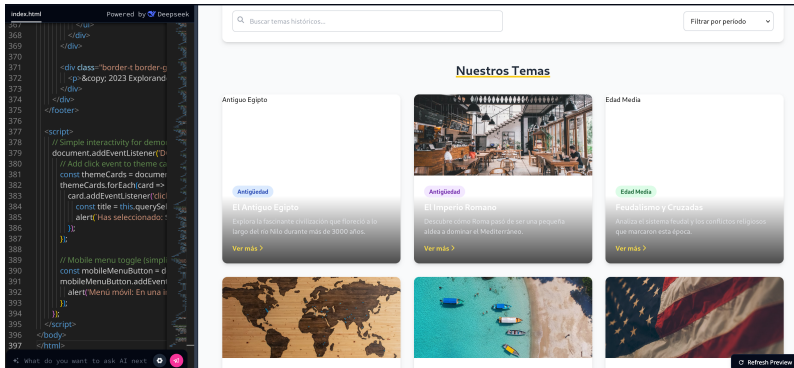
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# Example: DeepSite



<https://huggingface.co/spaces/enzostvs/deepsite>

<https://huggingface.co/spaces/enzostvs/deepsite/discussions/74>

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# More examples

<https://huggingface.co/spaces/>

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# Generative AI in your computer

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# Speech to text: Whisper

```
#!/usr/bin/python3  
import whisper
```

```
model = whisper.load_model('tiny')  
transcription = model.transcribe('recording.wav')  
print(transcription['text'])
```

```
$ whisper speech.wav --language Spanish
```

<https://openai.com/blog/whisper/>  
<https://github.com/openai/whisper>  
License: MIT (open source)

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# Text to speech: Coqui TTS

```
$ tts --text "Texto" \  
--model_name tts_models/es/mai/tacotron2-DDC \  
--out_path speech.wav
```

<https://github.com/idiap/coqui-ai-TTS>

License: MPL-2.0 (open source)

# Stable Diffusion

EasyDiffusion:

<https://easydiffusion.github.io>

Fooocus:

<https://github.com/l1llyasviel/Fooocus>

AUTOMATIC1111:

[https://github.com/AUTOMATIC1111/  
stable-diffusion-webui](https://github.com/AUTOMATIC1111/stable-diffusion-webui)

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# Stable Diffusion (2)

SD.next:

<https://github.com/vladmandic/automatic>

ComfyUI:

<https://github.com/comfyanonymous/ComfyUI>

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# Text generation models

Llama4, Llama Community License

<https://llama.com/models/llama-4/>  
<https://github.com/meta-llama/>

DeepSeek, DeepSeek License

<https://deepseek.com/>  
<https://huggingface.co/deepseek-ai>

# Text generation models (2)

Mistral, Apache 2.0

<https://mistral.ai/>

<https://huggingface.co/mistralai>

Qwen, Apache 2.0

<https://huggingface.co/Qwen>

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# Leaderboard

| Rank (UB) | Model                                           | Arena Elo | 95% CI | Votes  | Organization | License           | Knowledge Cutoff |
|-----------|-------------------------------------------------|-----------|--------|--------|--------------|-------------------|------------------|
| 1         | <a href="#">GPT-4-Turbo-2024-04-09</a>          | 1258      | +3/-3  | 44592  | OpenAI       | Proprietary       | 2023/12          |
| 2         | <a href="#">GPT-4-1106-preview</a>              | 1252      | +2/-3  | 76173  | OpenAI       | Proprietary       | 2023/4           |
| 2         | <a href="#">Gemini 1.5 Pro API-0409-Preview</a> | 1249      | +3/-3  | 61011  | Google       | Proprietary       | 2023/11          |
| 2         | <a href="#">Claude 3 Opus</a>                   | 1248      | +2/-2  | 101863 | Anthropic    | Proprietary       | 2023/8           |
| 3         | <a href="#">GPT-4-0125-preview</a>              | 1246      | +3/-2  | 70239  | OpenAI       | Proprietary       | 2023/12          |
| 6         | <a href="#">Bard (Gemini Pro)</a>               | 1208      | +5/-6  | 12387  | Google       | Proprietary       | Online           |
| 6         | <a href="#">Llama-3-70B-Instruct</a>            | 1208      | +3/-3  | 75844  | Meta         | Llama 3 Community | 2023/12          |
| 7         | <a href="#">Reka-Core-20240501</a>              | 1199      | +4/-4  | 18735  | Reka AI      | Proprietary       | Unknown          |
| 8         | <a href="#">Claude 3 Sonnet</a>                 | 1200      | +2/-3  | 84252  | Anthropic    | Proprietary       | 2023/8           |
| 10        | <a href="#">GPT-4-0314</a>                      | 1189      | +3/-3  | 53446  | OpenAI       | Proprietary       | 2021/9           |
| 10        | <a href="#">Qwen-Max-0420</a>                   | 1186      | +5/-7  | 10500  | Alibaba      | Proprietary       | Unknown          |
| 10        | <a href="#">Command R+</a>                      | 1189      | +3/-3  | 50490  | Cohere       | CC-BY-NC-4.0      | 2024/3           |
| 12        | <a href="#">Claude 3 Haiku</a>                  | 1180      | +2/-3  | 74097  | Anthropic    | Proprietary       | 2023/8           |
| 13        | <a href="#">Qwen1.5-110B-Chat</a>               | 1172      | +7/-8  | 6019   | Alibaba      | Qianwen LICENSE   | 2024/4           |

<https://lmarena.ai/?leaderboard>

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# Inference engines

## llama.cpp

<https://github.com/ggerganov/llama.cpp>

## Ollama

<https://ollama.com/>

## vLLM

<https://github.com/vllm-project/vllm>

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# Chat frontends

Oobabooga:

[https://github.com/oobabooga/  
text-generation-webui](https://github.com/oobabooga/text-generation-webui)

openplayground:

<https://github.com/nat/openplayground>

catAI

<https://github.com/withcatai/catai>

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# Collections of consented, open data

Chatbot Arena Leaderboard

<https://chat.lmsys.org/>

LAION Open Empathic

<https://laion.ai/blog/open-empathic/>

LAION Datasets

<https://laion.ai/projects/>

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# Agent frameworks

## Camel–AI Owl

<https://github.com/camel-ai/owl><https://github.com/camel-ai>

## LangChain

<https://langchain.com/>

## Smolagents

<https://github.com/huggingface/smolagents>

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# Infrastructure to play, to share

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## Hugging Face

“GitHub for ML”

<https://huggingface.co>

The screenshot displays the Hugging Face homepage. At the top, there's a search bar and navigation links for Models, Datasets, Spaces, and Docs. Below the search bar, there are several sections:

- Tasks:** A grid of task categories including Image Classification, Translation, Image Segmentation, Fill-Mask, Automatic Speech Recognition, Token Classification, Sentence-Similarity, Audio Classification, Question Answering, Summarization, and Zero-Shot Classification.
- Libraries:** A section for popular libraries like PyTorch, TensorFlow, and JAX.
- Datasets:** A list of datasets such as mozilla-foundation/common\_voice\_T\_0, squad, wikipedia, common\_voice, glia, emotion, xtrame, and reilbert/semeval2012\_relational\_similarity\_v6.
- Languages:** A section for languages supported by the models, including English, French, Spanish, German, Chinese, Japanese, Portuguese, and Russian.
- Licenses:** A section for various licenses like apache-2.0, mit, and afi-3.0.
- Other:** A section for other resources like AutoTrain Compatible, Eval Results, and Carbon Emissions.

On the right side, there's a 'Models' section with a filter by name. It lists several models with their names, update dates, sizes, and download counts, including bert-base-uncased, gpt2, openai/clip-vit-large-patch14, distilbert-base-uncased-finetuned-sst-2-english, allenai/specter, distilbert-base-uncased, bert-base-multilingual-cased, Jean-Baptiste/camenbert-ner, and roberta-base.

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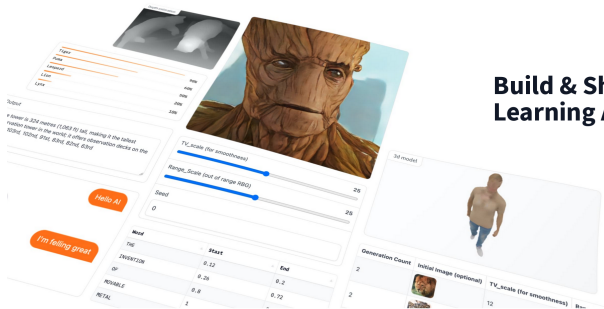
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# Gradio



## Build & Share Delightful Machine Learning Apps

Gradio is the fastest way to demo your machine learning model with a friendly web interface so that anyone can use it, anywhere!

Get Started

Star

10,955

<https://gradio.app/>  
License: Apache 2.0

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# Diffusers



Pretrained diffusion models (vision, audio, etc.)  
Modular toolbox for inference & training of  
diffusion models

<https://github.com/huggingface/diffusers>

License: Apache 2.0

# Model frameworks, etc

PyTorch

<https://pytorch.org/>

TensorFlow

<https://tensorflow.org/>

Keras

<https://keras.io/>

Cuda

[https://developer.nvidia.com/  
cuda-toolkit](https://developer.nvidia.com/cuda-toolkit)

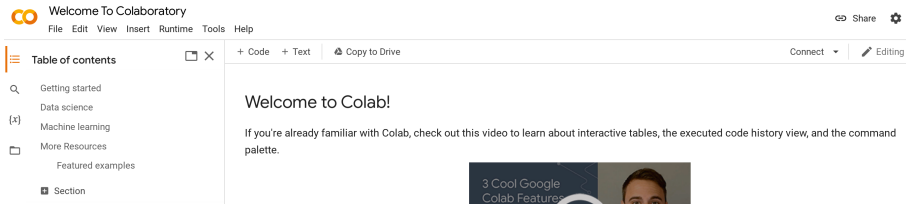
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## Collab



Python in the browser, zero configuration  
Access to GPUs & easy sharing

<https://colab.research.google.com/>

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# Jupyter



## Python in the browser, easy

<https://jupyter.org/>

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## API Providers

OpenRouter: <https://openrouter.ai/>Groq: <https://groq.com/>Together.ai: <https://together.ai>SambaNova <https://sambanova.ai/>

...and many more

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# Many issues raised

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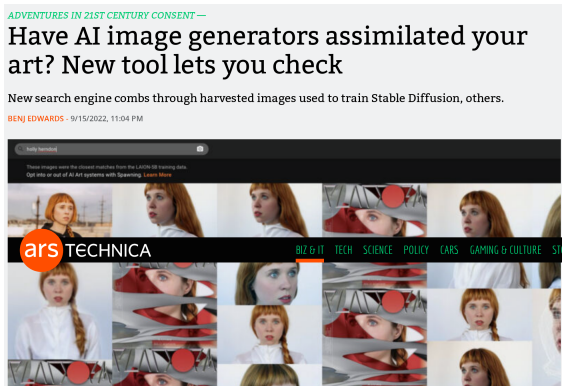
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# Intellectual property (training set)



<https://haveibeentrained.com/>

<https://arstechnica.com/information-technology/2022/09/have-ai-image-generators-as>

# Intellectual property (results)

Impact of Technology Deep Dive Report I

## STUDY ON THE IMPACT OF ARTIFICIAL INTELLIGENCE ON THE INFRINGEMENT AND ENFORCEMENT OF COPYRIGHT AND DESIGNS

[https://euipo.europa.eu/tunnel-web/secure/webdav/guest/document\\_library/observatory/documents/reports/2022\\_Impact\\_AI\\_on\\_the\\_Infringement\\_and\\_Enforcement\\_CR\\_Designs/2022\\_Impact\\_AI\\_on\\_the\\_Infringement\\_and\\_Enforcement\\_CR\\_Designs\\_FullR\\_en.pdf](https://euipo.europa.eu/tunnel-web/secure/webdav/guest/document_library/observatory/documents/reports/2022_Impact_AI_on_the_Infringement_and_Enforcement_CR_Designs/2022_Impact_AI_on_the_Infringement_and_Enforcement_CR_Designs_FullR_en.pdf)

# Intellectual property

- Can models be trained on anything public?
- Are models subject to copyright law?
- Who is the author of the production of a model?
- Can anybody besides the author claim rights on the production of a model

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## Licenses

| Model               | Model License                             | Description                                                                                                                                                            | Link to License                                                                                                                                                                       |
|---------------------|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GPT-2               | MIT License + generated output disclaimer | Permissive open source license                                                                                                                                         | <a href="https://github.com/openai/gpt-2/blob/master/LICENSE">https://github.com/openai/gpt-2/blob/master/LICENSE</a>                                                                 |
| GPT-3               | Exclusive                                 | Licensed to                                                                                                                                                            | Microsoft                                                                                                                                                                             |
| YOLO                | YOLO License                              | Public domain license                                                                                                                                                  | <a href="https://github.com/pjreddie/darknet/blob/master/LICENSE">https://github.com/pjreddie/darknet/blob/master/LICENSE</a>                                                         |
| DALLE-pytorch       | MIT License                               | Pytorch implementation of DALLE created by individual researcher                                                                                                       | <a href="https://github.com/lucidrains/DALLE-pytorch/blob/main/LICENSE">https://github.com/lucidrains/DALLE-pytorch/blob/main/LICENSE</a>                                             |
| Stable Diffusion    | CreativeML Open RAIL-M                    | Open & Responsible AI License (RAIL) created by Stability.ai and adapted from the BLOOM RAIL license, including use-based restrictions (see attachment A)              | <a href="https://huggingface.co/spaces/CompVis/stable-diffusion-license">https://huggingface.co/spaces/CompVis/stable-diffusion-license</a>                                           |
| OPT                 | OPT-1.75B License                         | Meta restrictive license enabling use of the model weights for research purposes while establishing a set of use-based restrictions, which could be considered a RAIL. | <a href="https://github.com/facebookresearch/metaseq/blob/main/projects/OPT/MODEL_LICENSE.md">https://github.com/facebookresearch/metaseq/blob/main/projects/OPT/MODEL_LICENSE.md</a> |
| BigScience          | BigScience OpenRAIL-M                     | Open & Responsible AI License (RAIL) created by BigScience and adapted from the BLOOM RAIL license, including use-based restrictions (see attachment A)                | <a href="https://huggingface.co/spaces/bigscience/license">https://huggingface.co/spaces/bigscience/license</a>                                                                       |
| Tsinghua University | GLM-130B license                          | Restrictive license enabling use of the model weights for research purposes                                                                                            | <a href="https://github.com/THUDM/GLM-130B/blob/main/MODEL_LICENSE">https://github.com/THUDM/GLM-130B/blob/main/MODEL_LICENSE</a>                                                     |

<https://hackmd.io/@jending12/HyvMU8sJo><https://thegradient.pub/>[machine-learning-ethics-and-open-source-li](https://machine-learning-ethics-and-open-source-li)

## Open Generative AI

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What is an “open”  
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Generative AI in  
your computer

Infrastructure to  
play, to share

Many issues raised

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# Bias



Mugshot of a technical speaker, machine learning expert,  
smiling, long hair, big eyes [t-shirt, curly hair]

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# Security



[Privacy](#) [Careers](#) [Disclosure Policy](#) [Technical Advisories](#) [Public Reports](#) [2021 Research Report](#)

[Contact Us](#)

## Whitepaper – Practical Attacks on Machine Learning Systems

 Jennifer Fernick

 Machine Learning, Offensive Security & Artificial Intelligence, Research, Research Paper, Whitepaper

 July 6, 2022  1 Minute

*Written by Chris Anley, Chief Scientist, NCC Group*

[https://research.nccgroup.com/2022/07/06/  
whitepaper-practical-attacks-on-machine-learning-systems/](https://research.nccgroup.com/2022/07/06/whitepaper-practical-attacks-on-machine-learning-systems/)  
<https://simonwillison.net/2022/Sep/12/prompt-injection/>

# Impact on professionals

- No more draw for hire as a profession?
- New opportunities for artists?
- Access to models as a fundamental need?

Is this different from the invention of photography?

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# Prompt engineers

A new profession

Artists, engineers, craftsmen?

Is it here to stay?

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# What is true?

## Make-A-Video

Make-A-Video research builds on the recent progress made in text-to-image generation technology built to enable text-to-video generation. The system uses images with descriptions to learn what the world looks like and how it is often described. It also uses unlabeled videos to learn how the world moves. With this data, Make-A-Video lets you bring your imagination to life by generating whimsical, one-of-a-kind videos with just a few words or lines of text.



<https://makeavideo.studio/>

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# The future just started

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# References, credits, license

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- Transformers-Tutorials

<https://github.com/NielsRogge/Transformers-Tutorials>

- Vision Transformers

<https://cameronrwolfe.substack.com/p/vision-transformers>

- A walk through latent space with Stable Diffusion

[https://keras.io/examples/generative/random\\_walks\\_with\\_stable\\_diffusion/](https://keras.io/examples/generative/random_walks_with_stable_diffusion/)

- How Open Source is eating AI

<https://lspace.swyx.io/p/open-source-ai>

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- Awesome Diffusion Models

<https://github.com/heejkoo/Awesome-Diffusion-Models>

- /r/StableDiffusion at Reddit

<https://www.reddit.com/r/StableDiffusion>

- Open LLMs

<https://github.com/eugeneyan/open-llms>

- The Generative Landscape (WiP course)

<https://johnowhitaker.github.io/tgllcourse/>

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